

WEEK 1

Task1

Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that

1. Accept a Linux command as input.
2. Create a child process using fork().
3. Execute the command in the child process using an appropriate exec() system call.
4. Allow the parent process to wait for the child using wait ().
5. Display the Process ID (PID) of both parent and child processes.

Source Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main(){
    char command[100];
    pid_t pid;
    printf("Enter a Linux command: ");
    scanf("%s", command);
    pid = fork();
    if(pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }
    else if(pid == 0) {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());

        execlp(command, command, NULL);
        printf("Command execution failed.\n");
        exit(1);
    } else {
        wait(NULL);
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child process completed.\n");
    }
    return 0;
}
```

Output:

```
Enter a Linux command: ls

Child Process
Child PID : 437
OSSP code.c files.c gm.c hello.c os program.c programming.c source_code
code files gm hello operating os.c programming programming.c.save source_code.c

Parent Process
Parent PID : 436
Child process completed.
```

Task2

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Report

1. Uname: Displays operating system and kernel information.

```
root@DESKTOP-VMN932K:~# uname
Linux
```

2. Lscpu: Displays CPU information.

```
root@DESKTOP-VMN932K:~# lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 4
On-line CPU(s) list:    0-3
Vendor ID:              GenuineIntel
Model name:             Intel(R) Core(TM) i3-8145U CPU @ 2.10GHz
CPU family:             6
Model:                  142
Thread(s) per core:     2
Core(s) per socket:     2
Socket(s):              1
Stepping:               12
BogoMIPS:               4608.01
```

3. Lsblk: Displays storage devices.

```

root@DESKTOP-VMN932K:~# lsblk
NAME
    MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /mnt/wslg/distro
/

```

4. ps: Displays currently running processes.

```

root@DESKTOP-VMN932K:~# ps
  PID TTY          TIME CMD
  476 pts/0    00:00:00 bash
  505 pts/0    00:00:00 ps

```

5. top: Displays real-time system performance.

```

root@DESKTOP-VMN932K:~# top
top - 13:56:44 up 22 min,  1 user,  load average: 0.06, 0.03, 0.00
Tasks: 22 total,  1 running, 21 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  7880.6 total,  7463.6 free,   438.7 used,   123.2 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7441.9 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   21680  13140  9744 S   0.0   0.2   0:00.74 systemd
    2 root        20   0    3180   2200  2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    9 root        20   0    3196   2128  2008 S   0.0   0.0   0:00.00 init
   64 root       19  -1   34052  12556  11488 S   0.0   0.2   0:00.25 systemd-journal
  112 root        20   0   25548   6924   5168 S   0.0   0.1   0:00.26 systemd-udev
  123 systemd+    20   0   21344  13252  11124 S   0.0   0.2   0:00.12 systemd-resolve
  124 systemd+    20   0   91036   7992   7024 S   0.0   0.1   0:00.10 systemd-timesyn
  168 root        20   0    4244   2688   2432 S   0.0   0.0   0:00.00 cron
  169 message+    20   0    9540   5432   4736 S   0.0   0.1   0:00.09 dbus-daemon
  176 root        20   0   17980   8888   7848 S   0.0   0.1   0:00.08 systemd-logind
  178 root        20   0  1756104  13024  10888 S   0.0   0.2   0:00.14 wsl-pro-service

```